Vlaanderen
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GPFS Parallel File System and Data Management

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Innovative Computing for A Smarter Flanders

Overview

- GPFS
  - Specs
  - Recap: what is a parallel file system
  - GPFS architecture
  - GPFS vs Lustre
- Data Management
  - Data management on two clusters
  - GPFS storage types
  - Data management policies on Mindwell
- External data storage
  - Research data platforms
  - Data transfer clients

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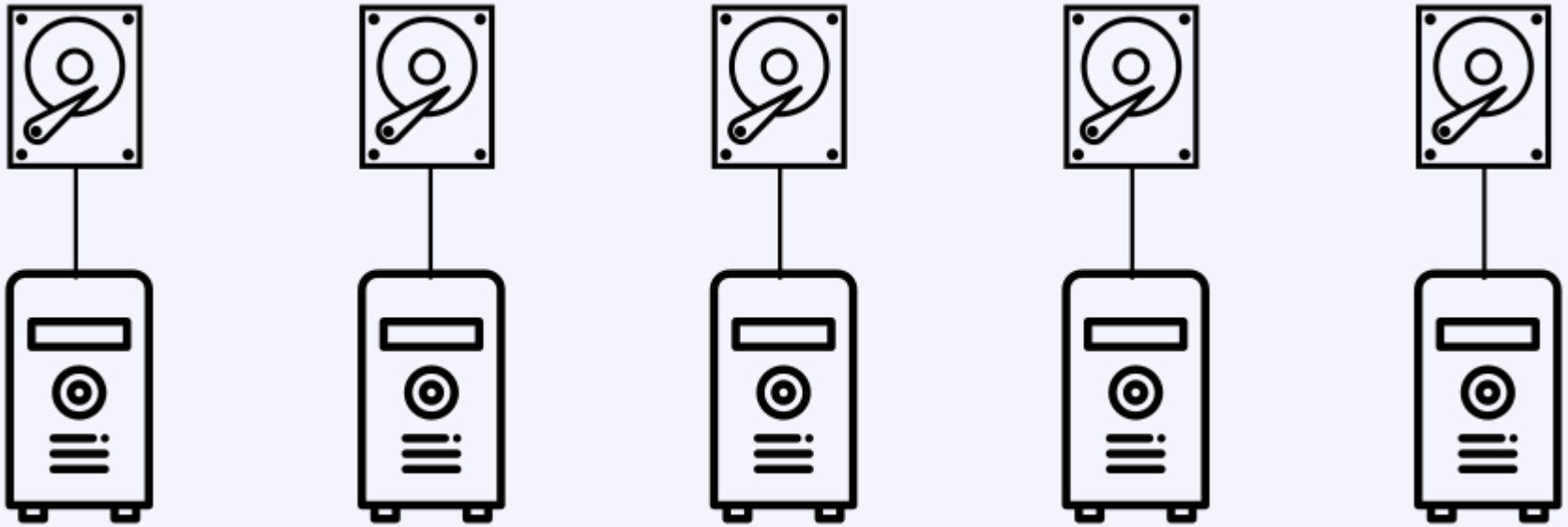
GPFS

- IBM's **G**eneral **P**arallel **F**ile **S**ystem or IBM Spectrum Scale
  - distributed file system
  - Allows parallel I/O
  - Commercial license
- Technical specifications
  - Storage
    - 4x hard disk enclosures each with 102 x 22 TB disks
    - Total of **5.8 PB useable storage**
  - Bandwidth (to be tested)
    - aggregated 40 GB/s read
    - aggregated 34 GB/s write
  - Connected to the cluster with InfiniBand NDR 400
    - 400 Gib/s

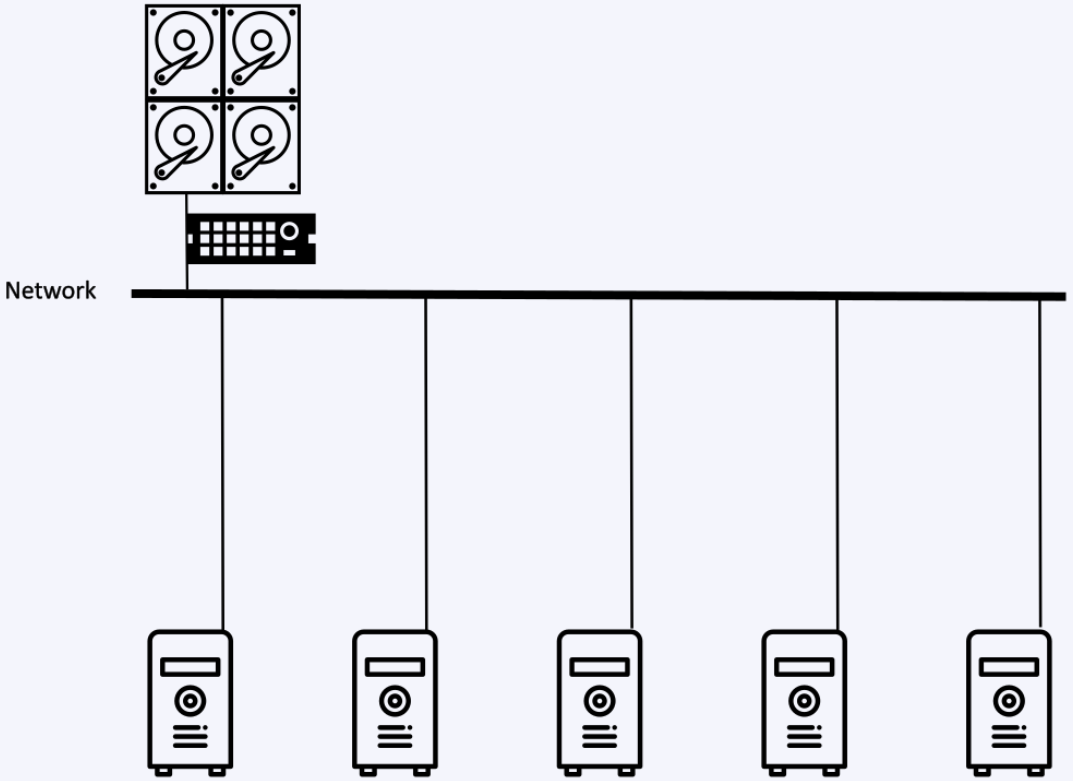
Recap: what is a parallel filesystem

Step 1: serial filesystem (e.g. node scratch)



Recap: what is a parallel filesystem

Step 2: shared filesystem (e.g. $VSC_DATA, /apps)



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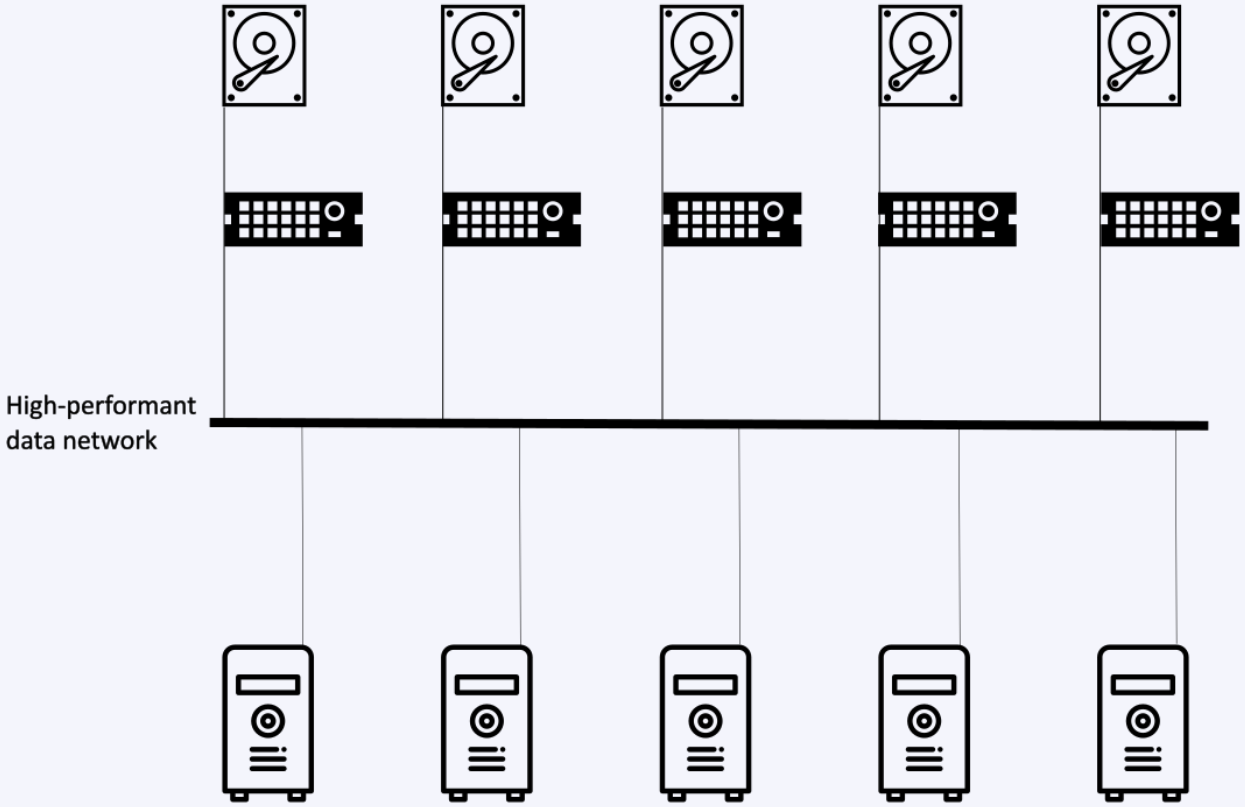
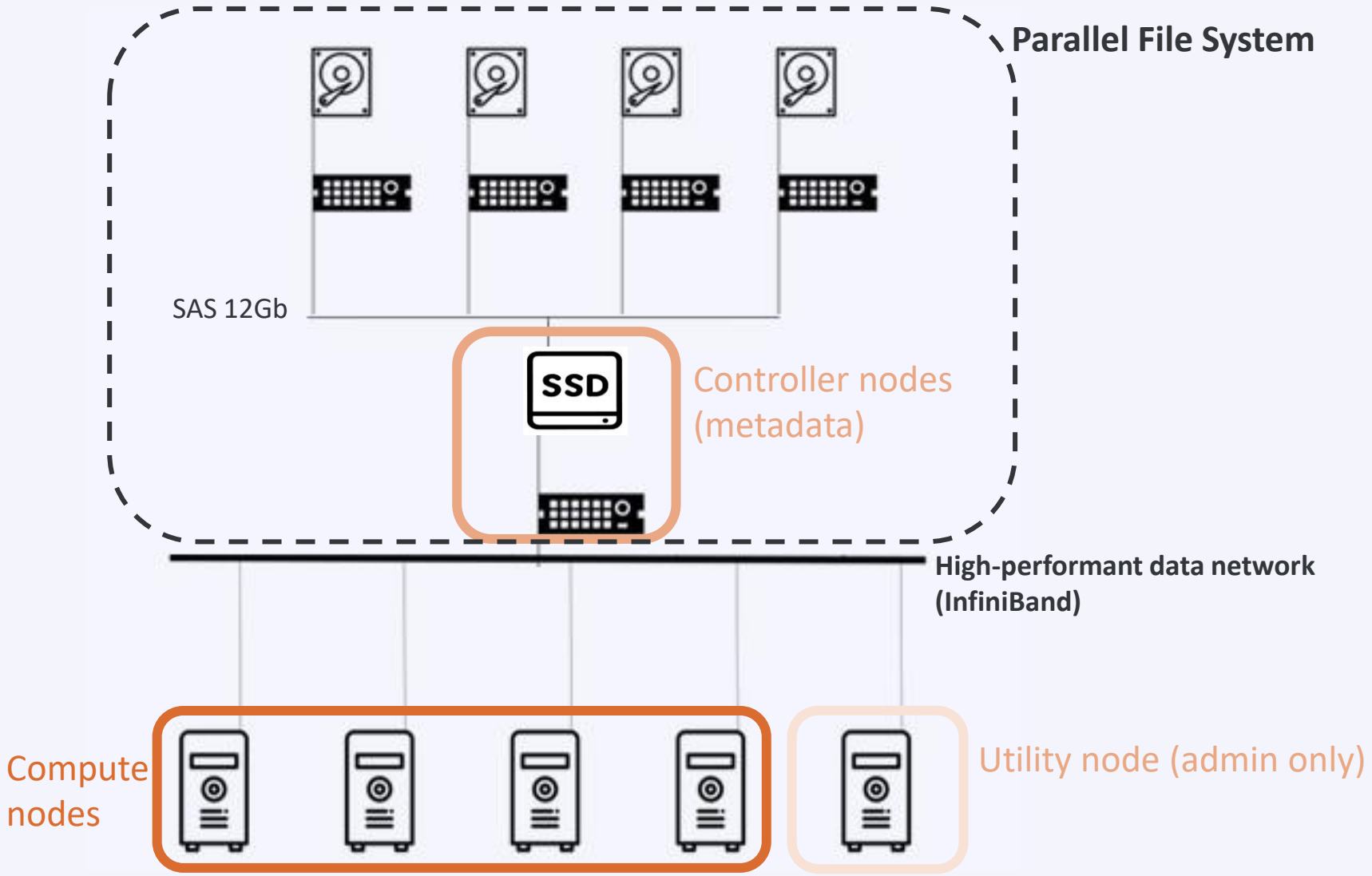
Recap: what is a parallel filesystem

Step 3: parallel filesystem (e.g. $VSC_SCRATCH, staging)



GPFS architecture



GPFS vs Lustre

	GPFS	Lustre
File striping	Completely managed by admins	Can be controlled by user
Workload suitability	Better performance across varying I/O patterns	Better performance for large sequential operations (e.g. simulations producing very large files)
License	Commercial	Open Source
Documentation	Only internal	Open source but requires expertise
Snapshots	Possible, but will not for $VSC_SCRATCH Considering for project storage	Not possible

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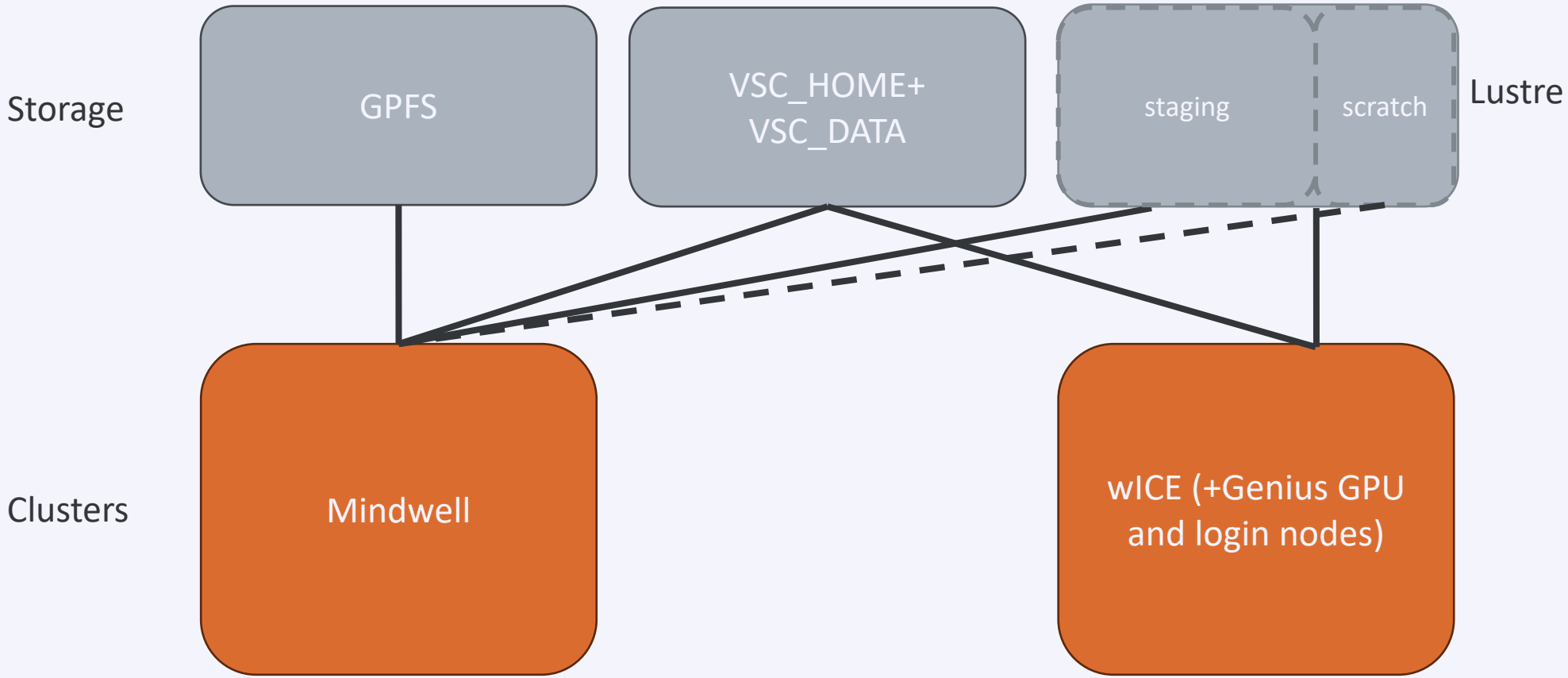
Data Management on two clusters



In place or confirmed

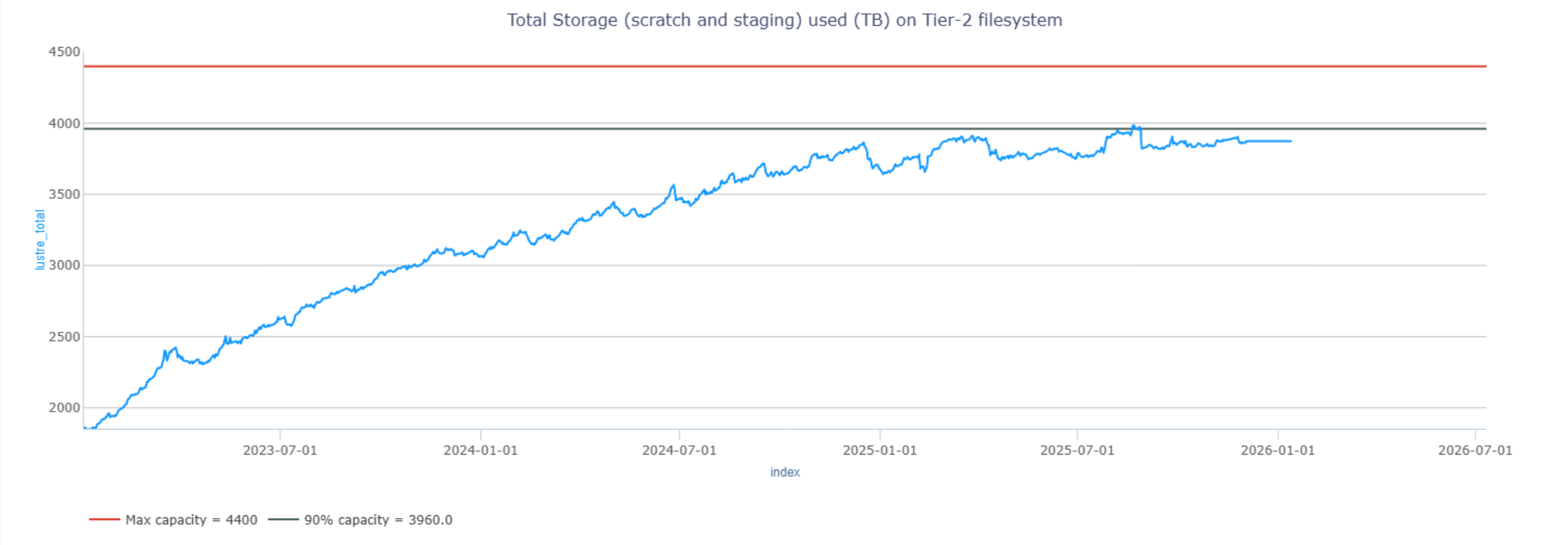
To be decided

GPFS storage types

	VSC_SCRATCH	Project storage
Path	/scratch/leuven/<vsc_prefix>/<vsc_id>	/project/leuven/prj_<id>
Size	500GB default + temporary increases upon request	Upon request. Max 10% of total GPFS storage per PI
Price	Free	20 euro per TB per year. Considering price increase for inactive data > 2 years
Permissions	Limited to user	Planned: extended permissions (ACLs) for moderators by default
Data removal	Data inactive > 30 days	None, but outflow strategy is necessary!

Data management policies on Mindwell

- To avoid filling up the project storage



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Data management policies on Mindwell

- Project storage is meant the **active computation phase**
  - Data actively being computed on
  - Data that will be used in the foreseeable future
- Explicit data outflow strategy is required for large requests (+ 100TB)
  - Required to meet with the support team
  - See our data management guidelines on https://docs.vscentrum.be/data/data_management_guidelines.html
  - e.g. outflow to external storage (ManGO, Tier-1 Data…)
- Software in project storage
  - Still allowed
  - Be careful with conda
    - Use alternatives
    - Use conda containers to limit file count
      - Do NOT use Lustre conda envs on Mindwell!

Data management on project storage: best practices

- Move data to external storage as fast as you possibly can
  - If not needed in the next 3-6 months, move it out
  - Use automatic transfers if possible
  - Follow-up within team
- Make use of the tools we provide
  - ACLs for moderator group
  - duduckdb for detailed storage overview

```
$ module load duduckdb
$ duduckdb /data/leuven/public/staging-stats/<stg_dir>.parquet --human-readable –
older-than=2024-03-26

directory:                size          inodes
=========================================================================.
:                         902.9GiB      771.7Ki
.ipynb_checkpoints:       -             -
dir1 :                    -             -
dir2:                     -             -
dir3:                     889.4GiB      746.7Ki
dir4:                     13.0GiB       25.0Ki
dir5:                     -             -
dir6:                     -             1.0
dir7:                     -             -
dir8:                     577.9MiB      41.0
dir9:                     -             -
```

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Research data platforms

- For KU Leuven researchers
  - request via ICTS service catalog

See https://rdm-docs.icts.kuleuven.be/mango/

- For HPC users
  - Project-based application procedure

VSC Tier-1 Data

See https://www.vscentrum.be/data

Data transfer clients

Existing tools

iCommands

Python iRODS client

ManGO Portal

globus

New/upcoming tools

Package for programming language ‘R’

Iron

User day survey

https://hpcleuven.limesurvey.net/621354